

MODEL (min player)

Group inputs

group 1: $x \in X_1$



S_1 : 3 measurements

group 2: $x \in X_2$



S_2 : 5 measurements

$\vdots$

group G: $x \in X_G$



S_G : ... measurements

Per-group encoders

ϕ_1

ϕ_2

ϕ_G

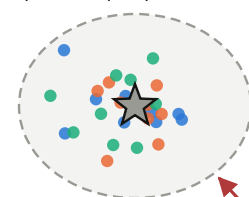
no shared parameters

Shared latent space

$Z = \mathbb{R}^d$

$\mu_1 = N(m_1, \Sigma_1)$

learnt class anchors



$\mu_2 = N(m_2, \Sigma_2)$

Shared head

ψ

one for all

$f_g = \psi \circ \phi_g$

Per-group losses

$L_1(\theta)$

$L_2(\theta)$

$L_G(\theta)$

cross-entropy, per group

LATENT-SPACE LOSSES

Alignment

$$L_{\text{align}} = \sum_{g,c} W_2^2(v_{g,c}, \mu_c)$$

each group's class cloud $\rightarrow$ its anchor

Separation

L_{sep} : keeps μ_c and $\mu_{c'}$ apart

anchors of different classes

Closed form for diagonal Gaussians:

$$W_2^2 = \|m_1 - m_2\|^2 + \|\sigma_1 - \sigma_2\|^2$$

GROUP WEIGHTS (max player)

Reference (offline)

$\tilde{R}_g$: best loss group g can reach alone

Excess

$$\Delta_g = L_g - \tilde{R}_g$$

signed, per group

Weight update

$$\lambda_g \propto \lambda_g \exp(\gamma \Delta_g)$$

on the simplex

Training objective

$$\min_{\theta} \sum_g \lambda_g L_g(\theta) + \lambda_{\text{fit}} L_{\text{align}} + \lambda_{\text{sep}} L_{\text{sep}}$$

gradients to encoders, head, anchors $\rightarrow$